

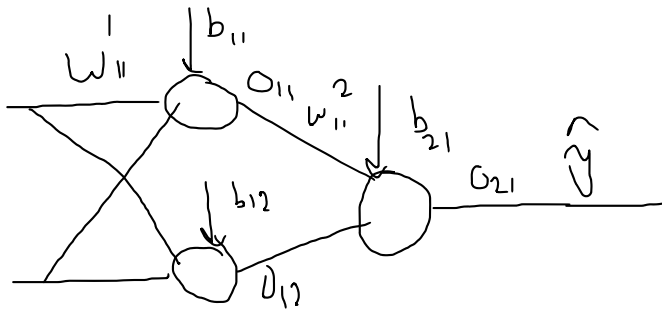
## Recap

20 December 2025 21:37

Loss func =  $f$  (all trainable params) ✓

$$L = f(\hat{y}) = g(w, b) \\ = f(g(w, b))$$

$\Rightarrow$  Sigmoid AF      for deep neural network  
 $\rightarrow$  Tanh AF



$$w'_{11} = 1$$

$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_{21}} \cdot \frac{dO_{21}}{dz_{21}} \cdot \frac{dz_{21}}{dO_{11}} \cdot \frac{dO_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw'_{11}}$$

individual derivatives values =  $\boxed{< 1}$

0.1      0.1      0.1      0.1      0.1      0.1

$$\frac{dL}{dw'_{11}} = 0.000001$$

$$w_n = w_0 \cdot \eta \frac{dL}{dw_0}$$

$$\eta = \boxed{< 1}$$

$$w'_{11} = w'_{11} - \eta \frac{dL}{dw'_{11}}$$

$$\begin{aligned}
 &= 1 - 0.1 \times 0.000001 \\
 &= 1 - 0.0000001 \\
 &= 0.9999999 \approx 1
 \end{aligned}$$

$$\sigma(w'_{11}) \approx \sigma(w_{11})$$

$$\boxed{50\% \text{ VGP}} = \text{Training}^X$$

Solutions?

1. Shallow Neural Network.
2. change Act. func
3. Proper weight initialization
4. Batch Normalisation.
5. Residual network.